

BENTWATERS, United KingdomWMO: **035963**Lat: **52.133N**Lon: **1.433E**Elev: **85**StdP: **14.65**Time Zone: **0.00 (EUW)**Period: **82-92**WBAN: **35048****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.4	28.4	16.0	12.4	28.4	19.3	14.5	29.8	33.9	48.7	30.2	46.7	5.9	90	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	76.7	64.5	73.7	63.3	71.7	62.2	66.6	74.5	64.9	71.8	63.5	69.7	9.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.5	69.1	62.3	84.3	68.4	60.8	79.8	67.2	31.1	74.0	29.9	71.5	28.9	69.8	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.2	17.9	DB	21.4	82.4	7.4	5.0	16.1	86.0	11.7	88.9	7.6	91.7	2.2	95.4	(4)
(5)			WB	19.5	69.5	6.9	1.6	14.5	70.7	10.5	71.6	6.6	72.5	1.5	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	40.3	39.4	42.7	46.3	52.4	58.4	63.3	62.5	59.0	53.0	46.2	43.7
	DBStd	9.69	6.37	6.07	4.77	4.98	4.46	4.73	4.20	4.26	4.33	5.11	5.80	5.31
	HDD50	1369	302	300	232	134	25	2	0	0	0	30	141	203
	HDD65	5301	765	718	692	560	390	204	86	103	186	373	564	660
	CDD50	1609	2	2	5	24	100	253	411	386	270	122	26	8
	CDD65	65	0	0	0	0	0	5	31	24	5	0	0	0
	CDH74	238	0	0	0	0	1	28	129	74	6	0	0	0
	CDH80	28	0	0	0	0	0	1	23	4	0	0	0	0
Wind	WSAvg	7.9	9.8	9.9	8.9	7.4	7.7	6.8	6.4	6.9	7.0	7.8	7.9	9.0
Precipitation	PrecAvg	23.5	2.0	1.6	1.5	1.5	1.8	2.1	2.0	2.3	1.9	2.5	2.2	2.2
	PrecMax	30.1	4.0	3.4	3.7	4.1	4.6	5.0	4.3	4.8	4.1	5.5	4.1	4.2
	PrecMin	16.9	0.3	0.3	0.2	0.0	0.2	0.4	0.4	0.2	0.2	0.2	0.6	0.7
	PrecStd	4.1	0.9	0.8	0.9	1.1	1.1	1.4	0.9	1.2	1.0	1.3	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	55.7	60.6	64.7	70.0	77.3	82.4	79.2	75.0	68.3	60.5	57.2
		MCWB	50.2	50.9	53.6	55.1	59.0	63.6	67.8	65.7	63.0	61.4	55.0	52.5
	2%	DB	53.5	53.4	55.3	61.0	65.9	73.4	77.1	75.5	71.7	64.5	57.6	55.3
		MCWB	48.7	49.2	49.1	53.2	57.0	62.8	65.4	64.2	62.3	58.2	52.8	51.2
	5%	DB	51.6	50.2	52.2	57.6	62.9	70.2	73.8	73.4	69.4	62.5	55.8	53.6
		MCWB	47.9	46.1	46.8	51.3	55.4	61.2	63.7	62.7	61.1	57.4	51.7	49.6
	10%	DB	48.6	48.0	50.2	55.2	60.9	67.7	71.6	71.2	66.4	60.8	55.1	51.9
		MCWB	45.6	44.4	45.5	49.7	53.6	59.5	62.4	62.1	59.5	55.7	51.1	48.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	51.6	53.5	57.2	60.0	67.0	68.9	67.8	65.7	62.3	55.2	53.2
		MCDB	54.5	55.7	60.5	63.7	67.3	74.7	78.3	75.5	71.4	67.8	58.7	56.3
	2%	WB	50.1	49.6	50.1	54.2	57.5	63.9	67.0	65.9	63.9	59.4	53.8	51.7
		MCDB	53.1	53.6	54.3	59.1	64.3	71.7	76.0	72.8	69.1	63.3	57.2	55.0
	5%	WB	48.1	46.6	48.2	51.9	56.2	61.7	65.2	64.4	62.3	57.5	52.6	50.2
		MCDB	50.7	49.6	52.5	57.9	62.5	69.1	72.5	71.1	67.0	61.6	56.0	53.1
	10%	WB	46.2	44.2	46.1	49.6	54.3	59.8	63.5	63.1	60.7	55.9	51.3	48.9
		MCDB	48.9	47.0	49.4	55.2	60.0	66.4	69.9	68.8	65.3	60.0	54.8	51.8
Mean Daily Temperature Range	5% DB	MDBR	8.4	7.8	10.8	12.0	12.7	13.2	13.7	14.1	12.8	10.7	9.3	8.2
		MCDBR	9.9	10.5	13.3	16.5	16.2	17.7	17.6	17.4	16.1	12.1	9.4	8.6
		MCWBR	9.8	9.2	10.2	10.9	10.3	10.5	9.8	9.2	10.6	8.9	8.4	7.8
	5% WB	MCDBR	9.4	10.5	13.3	16.2	16.0	16.9	16.4	16.0	13.6	11.6	8.8	8.3
		MCWBR	9.8	9.5	10.6	10.9	10.7	10.5	10.0	9.4	10.3	9.4	8.2	7.9
Clear-Sky Solar Irradiance	taub	0.328	0.346	0.382	0.403	0.409	0.412	0.411	0.401	0.380	0.362	0.334	0.319	
	taud	2.363	2.377	2.276	2.246	2.281	2.293	2.309	2.353	2.405	2.425	2.404	2.352	
	Ebn at Noon	211	243	254	263	268	267	266	261	253	233	207	192	
	Edh at Noon	20	25	34	39	40	40	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	237	451	804	1275	1535	1655	1578	1324	994	575	292	187	
	RadStd	22	67	93	129	114	128	156	97	81	53	32	23	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page